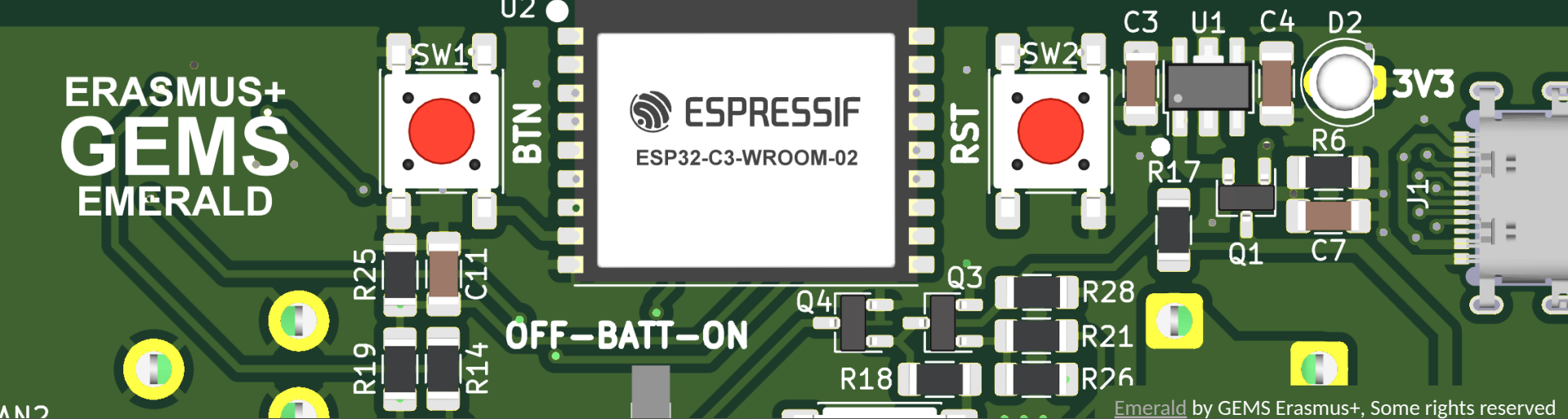


ERASMUS+
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EMERALD



Articulated robot

Building Robots: System Integration

Gašper Škulj

University of Ljubljana, Faculty of mechanical engineering



About

1 Articulated robot

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3 Forward kinematics

4 Inverse kinematics

Articulated robot

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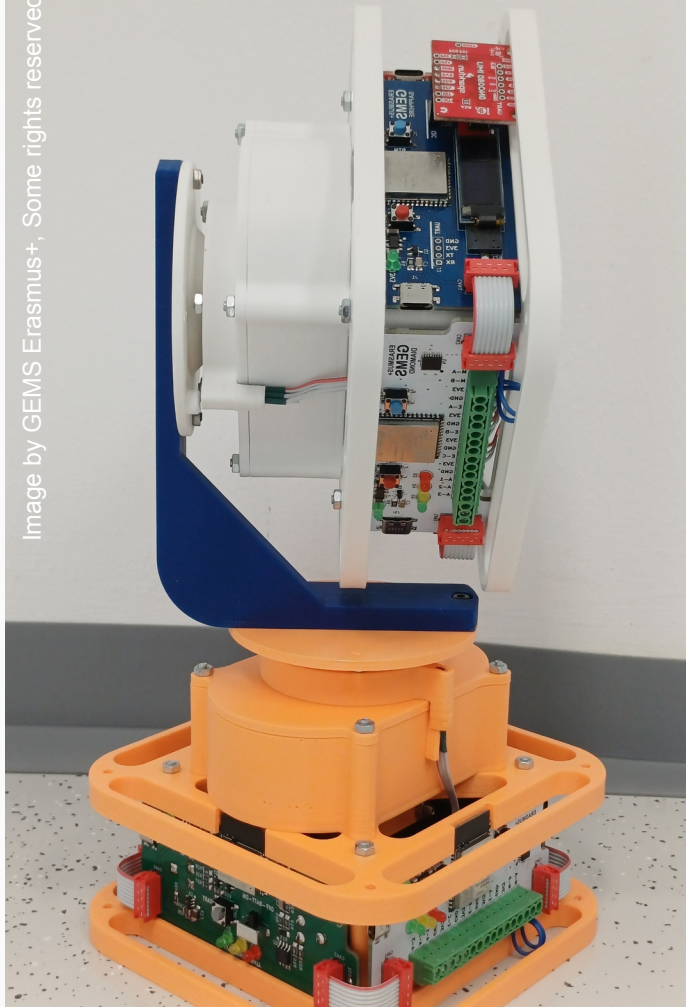


Robot type

- 1 Articulated robot → movement (to change pose)
- 2 1 DOF rotational/revolute joint link - rotational joint - link
- 3 Industrial robotic arm, quadruped robot, bipedal robot

Structure



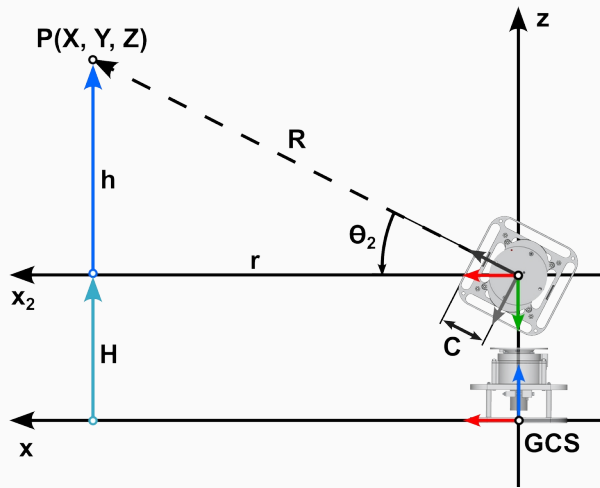


GEMS Articulated Robot

- 1 Fixed base + measuring head
2 rotational joints $\rightarrow$ 2 DOF
- 2 End effector: ultrasound sensor
measurement $\rightarrow$ third dimension
- 3 connecting piece (L shape) $\rightarrow$
accessible points on spherical shell

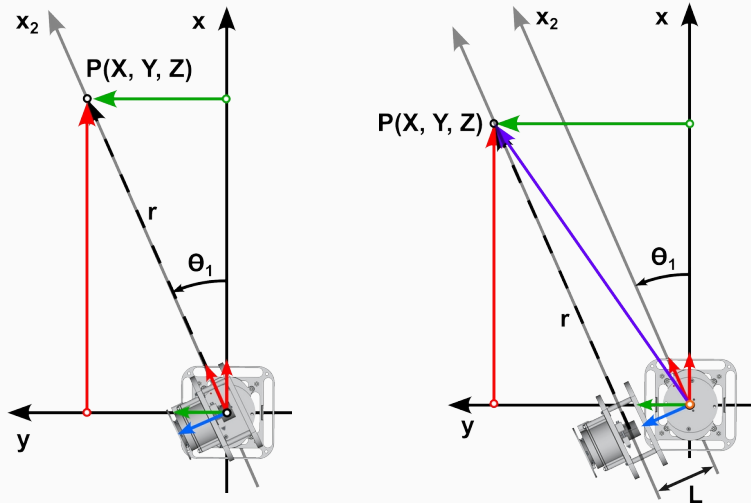
Kinematics





Forward kinematics

- 1 Measured position derived from joint angles and US measurement
- 2 Perpendicular axes of rotation
 $R = \text{ultrasound measurement} + C$
- 3 $\theta_1, \theta_2, R \rightarrow X, Y, Z$
 (offset H and L)

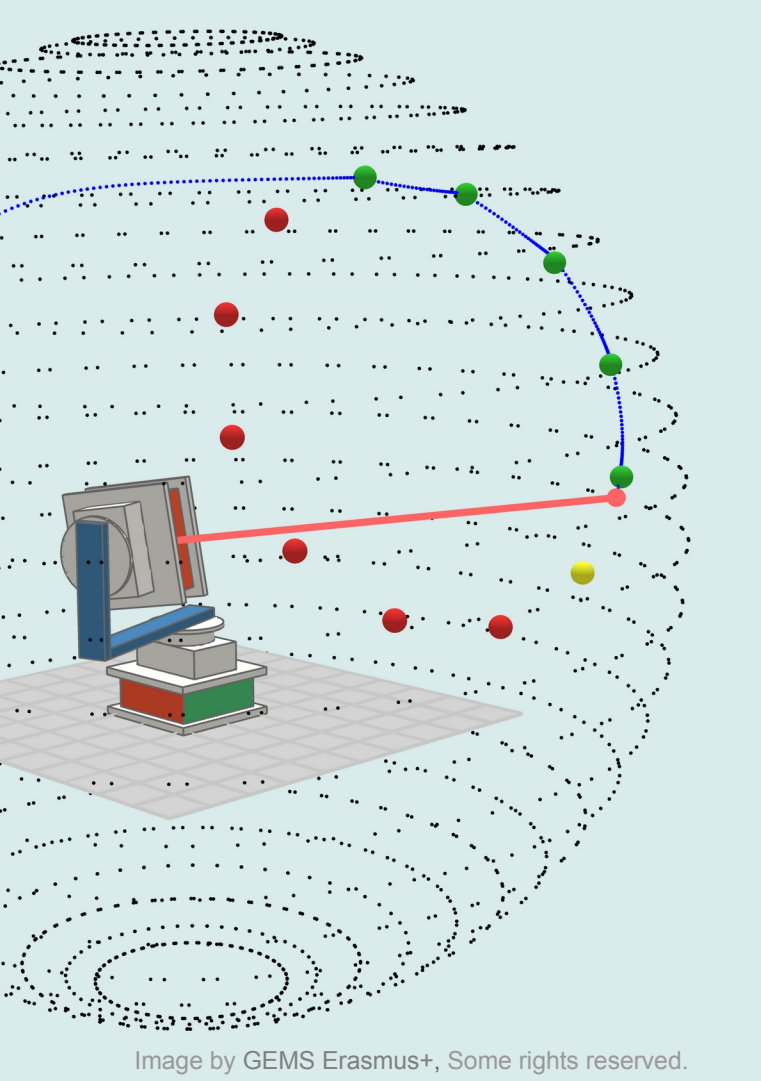




Inverse kinematics

- 1 Joint angles derived from the end-effector's desired position
- 2 $X, Y, Z \rightarrow \theta_1, \theta_2, (R)$
check $R \rightarrow$ end-effector dimension
- 3 It is possible there is no solution
workspace $\rightarrow$ spherical shell

Task



Forward kinematics

Arduino program

- 1 Target: ESP32C3/S3
reusable function
- 2 Input: joint angular positions,
offset L and measurement R
- 3 Output: X, Y, Z

Conclusion

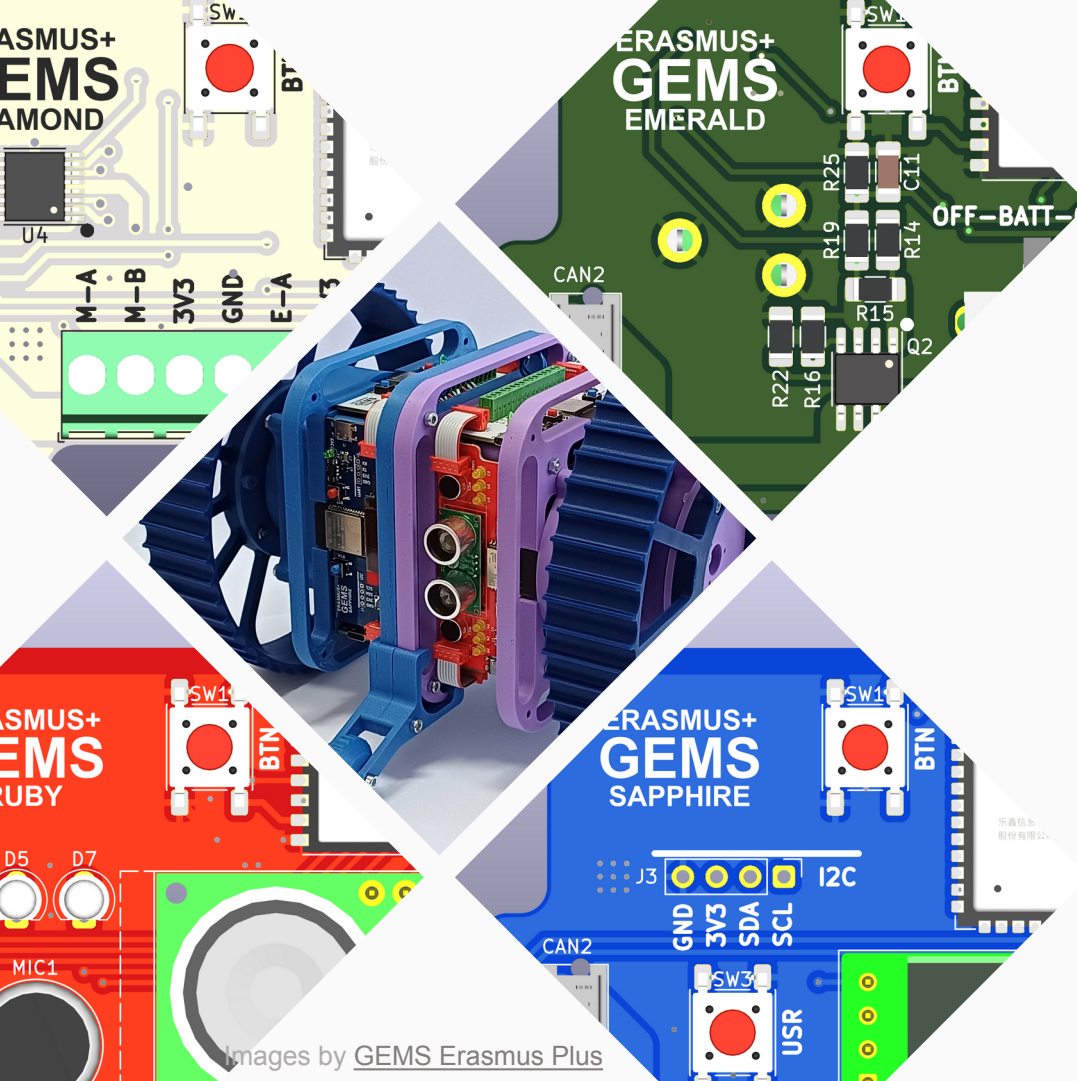




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Articulated Robot

- 1 Combination of 1 DOF rotational/revolute joints
- 2 Complexity → number and placement of joints
- 3 Robot geometry → kinematics difficulty level



Images by [GEMS Erasmus Plus](#)

Thank you for watching!

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**Co-funded by
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